A Safer, Better-Documented Network for Small Businesses

Project summary | Anurag Jassal

The problem

Many small businesses with more than one location run a single shared network for everything: staff computers, systems holding customer or client records, guest Wi-Fi, and phones. If a visitor's phone or one infected laptop connects, it can reach everything else. And when something breaks, there is often no map of how the network is set up, so fixing it takes longer and costs more.

What I built

A working model of a two-location business network (a headquarters and a branch), designed from written requirements. The example in the lab is a small clinic group, but the same design suits offices, professional firms, retail shops, and non-profits.

- **Separate zones.** Staff, servers, guests, and administration each get their own space, so a problem in one doesn't spread to the others.
- **Guests get internet only.** Visitors cannot reach internal systems.
- **Locked-down administration.** Only a dedicated admin zone can manage the network equipment, over encrypted logins.
- **Protected physical ports.** Unused network sockets are switched off, and each used socket accepts only its approved device.
- **Encrypted link between the sites, with a backup.** If the main connection fails, traffic automatically switches to a second, encrypted route.
- **In progress:** central staff accounts and security rules, plus a step-by-step procedure for adding and removing staff.

How I know it works

- I wrote the requirements first, then a test plan: each check states what should happen (for example, "a guest device must not reach internal systems") and what actually happened.
- Problems found along the way, such as a wrongly connected cable and a missing setting, are recorded in a troubleshooting log with the cause and fix.
- Everything is documented: the address plan, equipment settings, diagrams, and test results, so someone else could rebuild or maintain it.

What a business gets from this approach

- Lower risk from a single compromised device, because the network is divided and access is limited.
- Clear documentation, so repairs and changes are faster.
- A repeatable process for new staff and departures, so access is granted and removed consistently.

> **Please note:** this is a simulated lab built in Cisco Packet Tracer. It has not been deployed in a real business or tested on physical equipment, and it does not by itself make an organization compliant with privacy or industry regulations.

What's next

Finish the central-accounts work, run deliberate break-and-fix exercises, and adapt the approach to a real small business through a network review.

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